Week 15 Progress Report

Details

- Name of Student: Tirthak Girish Likhar
- PRN: 22070521041
- Name of Industry: SS Infotech Pvt. Ltd.
- **Reporting Week: (13/04/2026 to 17/04/2026)**
- Institute Mentor Name: Dr. Shreyas Hole
- Industry Mentor Name: Mr. Viraj Patle

Overview of the Week

- Week 15 focused on simulating a real-world network reconnaissance attack using Nmap port scanning from the attacker VM and detecting the activity through Wazuh by analysing UFW firewall logs. This extended the SOC lab from active response (Week 14) into the reconnaissance detection phase, representing the earliest stage of the attack lifecycle under MITRE ATT&CK.

- The simulation was executed by installing Nmap 7.94SVN on the Ubuntu 22.04 attacker VM (192.168.1.7) and running a TCP Connect scan (-sT) against the Wazuh Server at 192.168.1.6. UFW on the server was already active with a default deny policy, causing it to generate BLOCK entries in /var/log/syslog for each denied port, providing clear forensic evidence of the port scan activity.

- Detection was validated by examining /var/log/syslog for UFW BLOCK entries from 192.168.1.7 across multiple destination ports. A custom Wazuh detection rule (Rule 100003) was also added to /var/ossec/etc/rules/local_rules.xml to match the UFW BLOCK syslog pattern and tag the activity with MITRE ATT&CK T1046 Network Service Discovery.

Objectives of the Week

- To install Nmap on the Ubuntu 22.04 attacker VM and verify the installation.

- To simulate a TCP Connect port scan (-sT) from the attacker VM (192.168.1.7) targeting the Wazuh Server (192.168.1.6).

- To detect the port scan activity through UFW BLOCK log entries in /var/log/syslog on the Ubuntu Server.

- To write and deploy a custom Wazuh detection rule (Rule 100003) in local_rules.xml to match the UFW BLOCK syslog pattern and tag the detection with MITRE ATT&CK T1046.

- To verify detection evidence by capturing syslog output showing packets from 192.168.1.7 blocked across multiple ports.
- To map the simulation to MITRE ATT&CK Technique T1046 Network Service Discovery.

Key Accomplishments

- Successfully installed Nmap 7.94SVN on the Ubuntu 22.04 attacker VM using: sudo apt install nmap -y
- Executed a TCP Connect scan (nmap -sT 192.168.1.6) from the attacker VM scan completed with Nmap reporting open ports (22/tcp, 443/tcp) and filtered ports across Wazuh services.
- Confirmed UFW BLOCK entries in /var/log/syslog showing packets from 192.168.1.7 being denied across multiple destination ports providing forensic evidence of the port scan.
- Added custom Wazuh detection rule (Rule 100003, Level 10) to /var/ossec/etc/rules/local_rules.xml targeting the UFW BLOCK syslog pattern and tagged with MITRE T1046.
- Restarted Wazuh Manager (sudo systemctl restart wazuh-manager) to load the new rule successfully.
- Resolved SSH connectivity issue caused by UFW + iptables LOG rule conflict during troubleshooting, restoring full lab access.
- Completed the reconnaissance detection simulation the SOC lab now covers the full kill-chain from reconnaissance (T1046) through brute force (T1110), credential abuse (T1078), active response, file integrity monitoring, and custom detection rules.

Issues Faced & Resolutions – Week 15

1. Wazuh dashboard did not generate a dedicated alert for the Nmap scan

- **Issue:** Despite UFW BLOCK entries being present in /var/log/syslog confirming the port scan from 192.168.1.7, the Wazuh Dashboard did not surface a dedicated alert for Rule 100003 during the observation window.
- **Resolution:** Verified the simulation directly via /var/log/syslog UFW BLOCK entries from 192.168.1.7 across multiple ports confirmed the scan was executed and blocked. Screenshots of syslog output were captured as primary evidence. Custom Rule 100003 was added to local_rules.xml for future detection.

2. SSH from Windows host to Ubuntu Server timed out during troubleshooting

- **Issue:** Enabling UFW and adding an iptables LOG rule (with --log-prefix "PORTSCAN: ") during Week 15 troubleshooting caused SSH connections from the Windows host to 192.168.1.6 to time out.

- **Resolution:** Opened the Ubuntu Server VM console directly in VirtualBox and ran: sudo ufw allow 22/tcp && sudo ufw reload. Removed the iptables LOG rule using: sudo iptables -D INPUT -j LOG --log-prefix "PORTSCAN: " --log-level 4. SSH access was fully restored.

Tools Used – Week 15

Tool / Technology	Purpose
Nmap 7.94SVN	Network port scan simulation from attacker VM (Ubuntu 22.04)
UFW (Uncomplicated Firewall)	Firewall on Wazuh Server — generated BLOCK entries in syslog
Wazuh Manager v4.7.5	SIEM/XDR — log ingestion, custom rule evaluation, alerting
Wazuh Dashboard v4.7.5	Alert visualisation and event monitoring
VirtualBox 7.x	Hypervisor hosting all VMs on internal network
Ubuntu Server 24.04	Wazuh Server — monitored and logging VM (192.168.1.6)
Ubuntu 22.04	Attacker VM (192.168.1.7)

Conclusion of Week 15

Week 15 successfully added network reconnaissance detection to the SOC lab by simulating a port scan using Nmap and verifying detection through UFW firewall logs in syslog.

The UFW BLOCK entries confirmed that the scan from 192.168.1.7 was visible in the logging pipeline even without a dedicated Wazuh dashboard alert, demonstrating how real-world SOC analysts use firewall logs as first-line forensic evidence when SIEM alerts are not immediately generated.

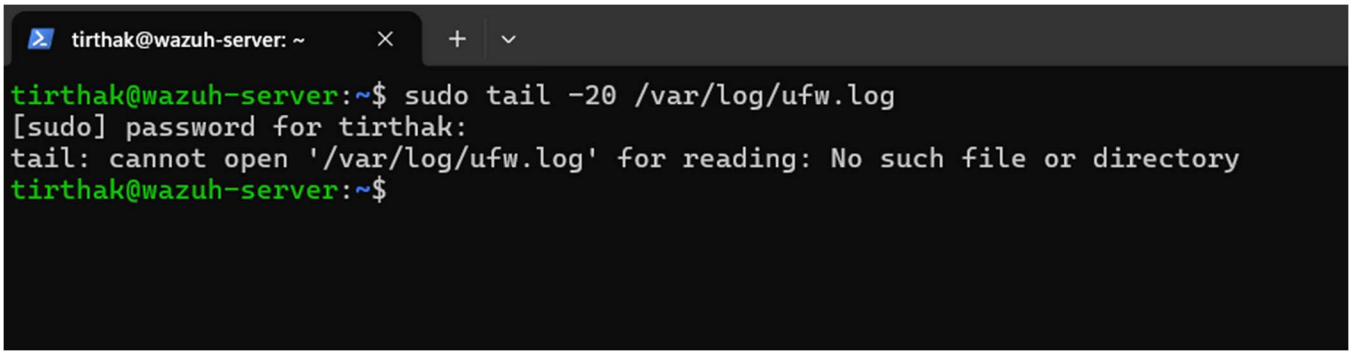
The addition of Custom Rule 100003 extends the Wazuh ruleset to natively detect and tag this pattern in future events. The lab environment now fully covers five simulation scenarios: Nmap reconnaissance (T1046), SSH brute force (T1110), failed login detection (T1078), file integrity monitoring, and automated active response — providing a comprehensive foundation for the final demo recording in Week 18.

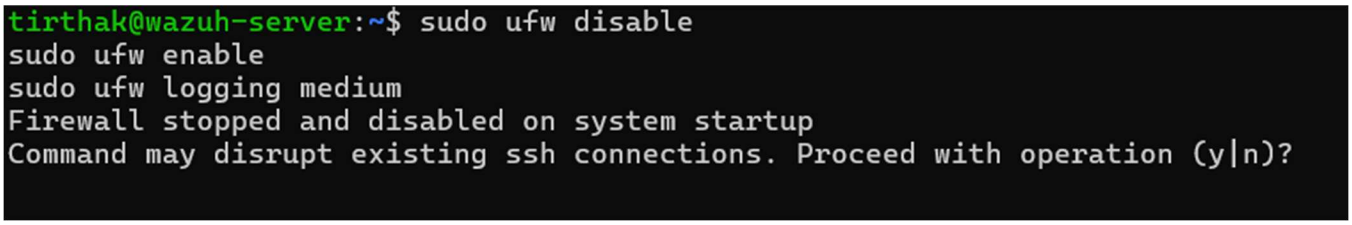
Port Scan Detection Summary – Week 15

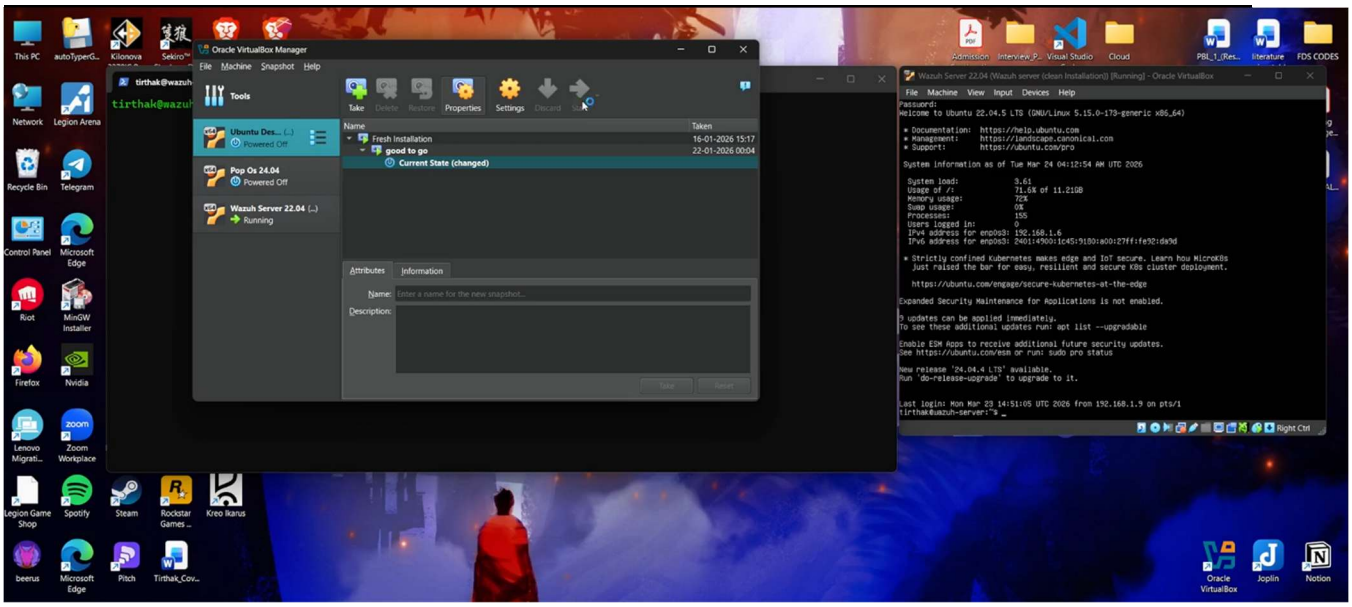
Parameter	Value
Detection Method	UFW Firewall Block — /var/log/syslog
Source IP	192.168.1.7 (Attacker VM — Ubuntu 22.04)
Target IP	192.168.1.6 (Wazuh Server — Ubuntu 24.04)
Scan Type	TCP SYN Scan (-sT)
Ports Targeted	Multiple (full-range Nmap scan)
Tool Used	Nmap 7.94SVN
MITRE Technique	T1046 — Network Service Discovery
Custom Rule	Rule 100003 (local_rules.xml)
Log Location	/var/log/syslog

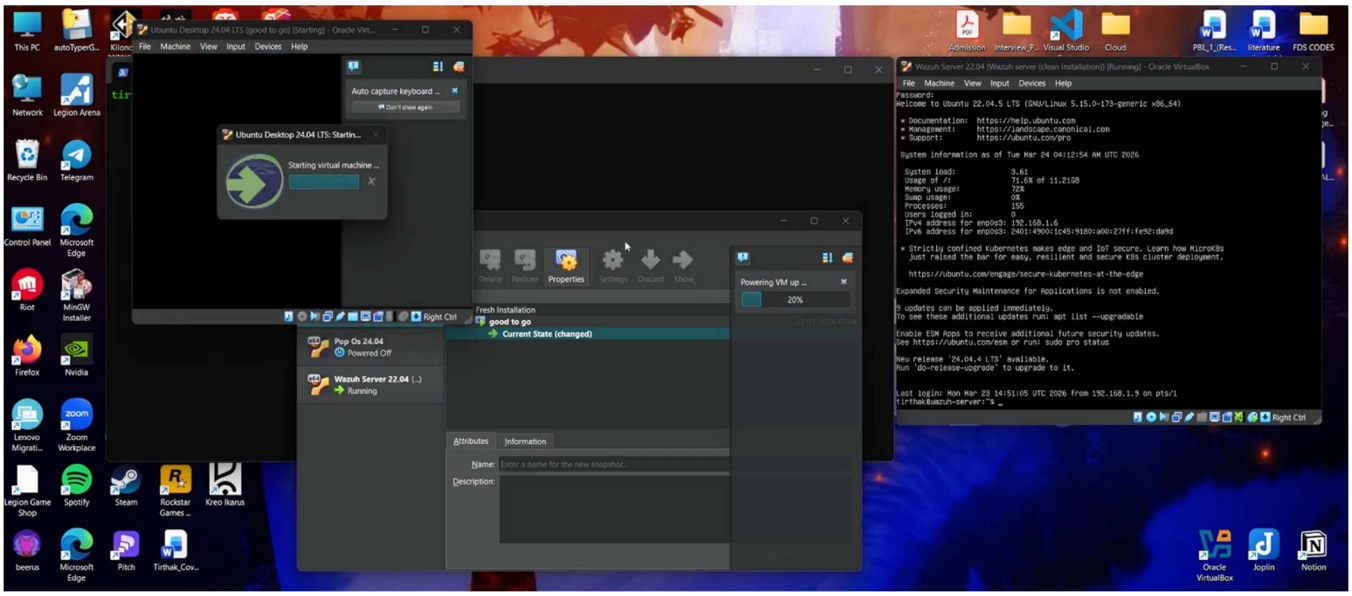
Screenshots of Activities (if Any)

[Nmap --version output, nmap -sT terminal output, UFW BLOCK entries in /var/log/syslog from 192.168.1.7, Rule 100003 in local_rules.xml, Wazuh Dashboard Threat Hunting view]






Oracle VirtualBox Manager
File Machine Snapshot Help
Tools
Take
Delete
Restore
Properties
Settings
Discard
Show
Ubuntu Des... (...)
Powered Off
Pop Os 24.04
Powered Off
Wazuh Server 22.04 (...)
Running
Name
Taken
Fresh Installation
16-01-2026 15:17
good to go
22-01-2026 00:04
Current State (changed)
Attributes
Information
Name:
Enter a name for the new snapshot...
Description:
Wazuh Server 22.04 (Wazuh server (clean Installation)) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-173-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Tue Mar 24 04:12:54 AM UTC 2026
System load: 3.61
Usage of /: 71.6% of 11.21GB
Memory usage: 72%
Swap usage: 0%
Processes: 155
Users logged in: 0
IPv4 address for enp0s3: 192.168.1.6
IPv6 address for enp0s3: 2401:4900:1c45:9180:a00:27ff:fe92:da9d
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.
https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Expanded Security Maintenance for Applications is not enabled.
9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Mon Mar 23 14:51:05 UTC 2026 from 192.168.1.9 on pts/1
tirthak@wazuh-server:~$
Right Ctrl


Ubuntu Desktop 24.04 LTS (good to go) [Starting] - Oracle Virt...
File Machine View Input Devices Help
Auto capture keyboard ...
Don't show again
Ubuntu Desktop 24.04 LTS: Startin...
Starting virtual machine ...
Right Ctrl
Delete
Restore
Properties
Settings
Discard
Show
Powering VM up ...
20%
Fresh Installation
good to go
Current State (changed)
Pop Os 24.04
Powered Off
Wazuh Server 22.04 (...)
Running
Attributes
Information
Name:
Enter a name for the new snapshot...
Description:
Wazuh Server 22.04 (Wazuh server (clean Installation)) [Running] - Oracle VirtualBox
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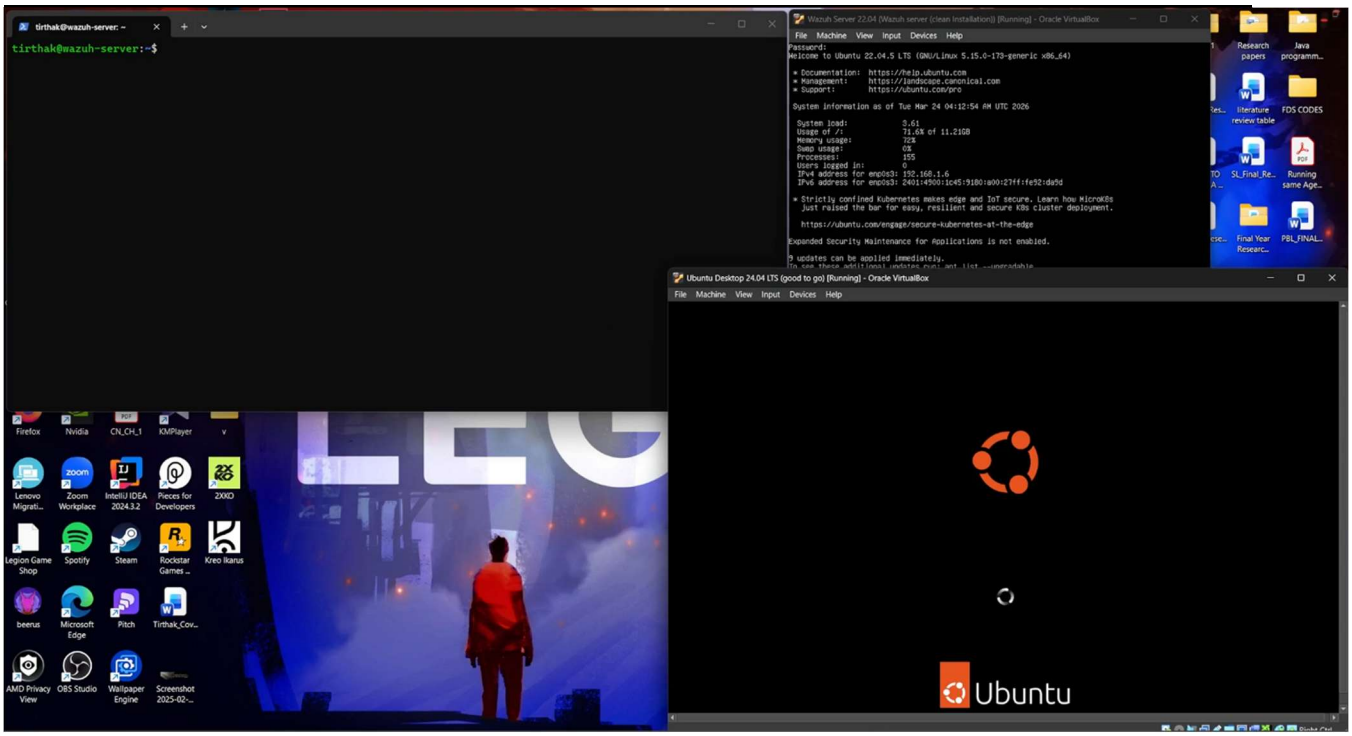
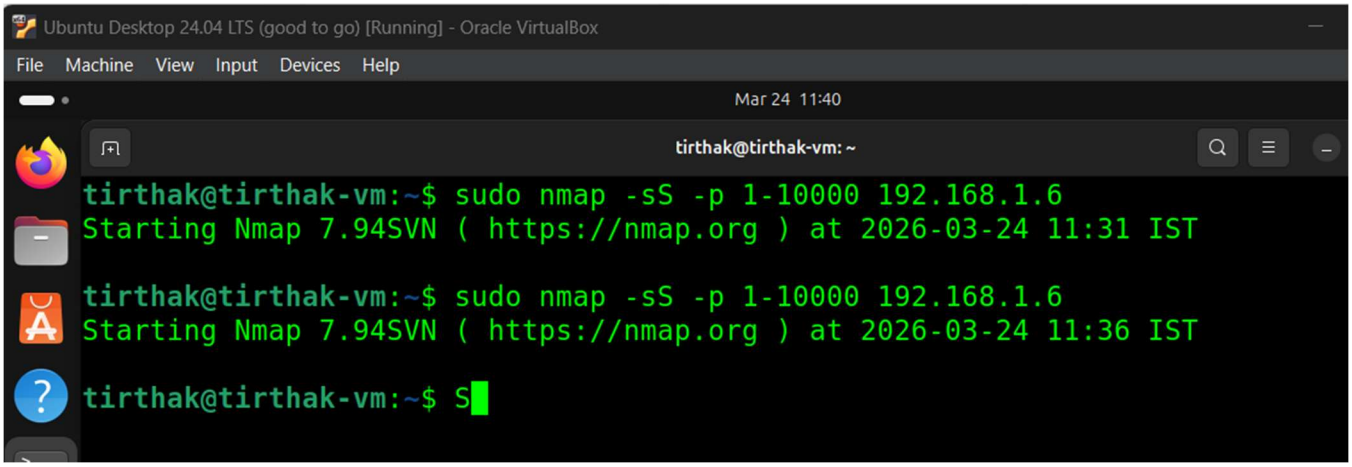
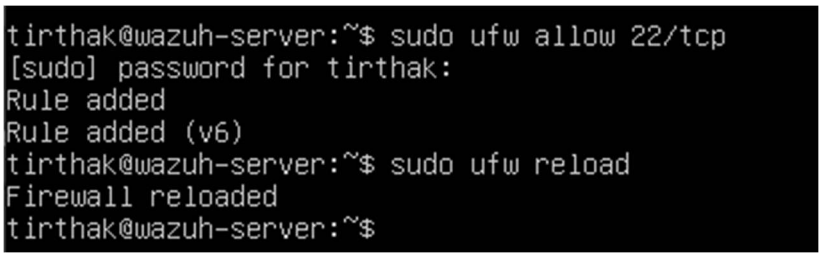
Usage of /: 71.6% of 11.21GB
Memory usage: 72%
Swap usage: 0%
Processes: 155
Users logged in: 0
IPv4 address for enp0s3: 192.168.1.6
IPv6 address for enp0s3: 2401:4900:1c45:9180:a00:27ff:fe92:da9d
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.
https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Expanded Security Maintenance for Applications is not enabled.
9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Mon Mar 23 14:51:05 UTC 2026 from 192.168.1.9 on pts/1
tirthak@wazuh-server:~$
Right Ctrl


tirthak@wazuh-server: ~
tirthak@wazuh-server:~$
Ubuntu Desktop 24.04 LTS (good to go) [Running] - Oracle VirtualBox
Ubuntu

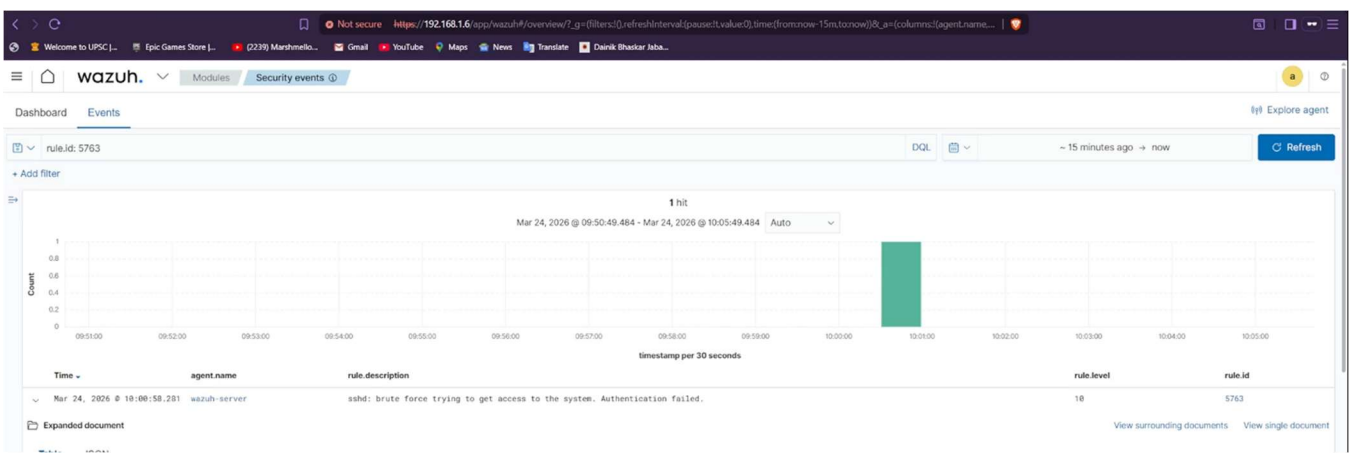

Ubuntu Desktop 24.04 LTS (good to go) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 24 11:40
tirthak@tirthak-vm: ~
tirthak@tirthak-vm:~$ sudo nmap -sS -p 1-10000 192.168.1.6
Starting Nmap 7.94SVN (https://nmap.org) at 2026-03-24 11:31 IST
tirthak@tirthak-vm:~$ sudo nmap -sS -p 1-10000 192.168.1.6
Starting Nmap 7.94SVN (https://nmap.org) at 2026-03-24 11:36 IST
tirthak@tirthak-vm:~$ S


tirthak@wazuh-server:~$ sudo ufw allow 22/tcp
[sudo] password for tirthak:
Rule added
Rule added (v6)
tirthak@wazuh-server:~$ sudo ufw reload
Firewall reloaded
tirthak@wazuh-server:~$

```
Last login: Tue Mar 24 05:41:16 2026 from 192.168.1.2
tirthak@wazuh-server:~$ sudo iptables -D INPUT -j LOG --log-prefix "PORTSCAN: " --log-level 4
[sudo] password for tirthak:
iptables: Bad rule (does a matching rule exist in that chain?).
tirthak@wazuh-server:~$ sudo iptables -L INPUT -n | head -10
Chain INPUT (policy DROP)
target     prot opt source               destination
ufw-before-logging-input  all  --  0.0.0.0/0            0.0.0.0/0
ufw-before-input  all  --  0.0.0.0/0            0.0.0.0/0
ufw-after-input  all  --  0.0.0.0/0            0.0.0.0/0
ufw-after-logging-input  all  --  0.0.0.0/0            0.0.0.0/0
ufw-reject-input  all  --  0.0.0.0/0            0.0.0.0/0
ufw-track-input  all  --  0.0.0.0/0            0.0.0.0/0
tirthak@wazuh-server:~$
```

```
tirthak@tirthak-vm: ~
tirthak@tirthak-vm:~$ sudo nmap -sT 192.168.1.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 11:47 IST
Nmap scan report for 192.168.1.6 (192.168.1.6)
Host is up (0.0058s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:92:DA:9D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 6.88 seconds
tirthak@tirthak-vm:~$
```




```
2026/03/24 04:33:59 active-response/bin/firewall-drop: Ended
```

```
2026/03/24 04:30:58 active-response/bin/firewall-drop: Aborted
2026/03/24 04:33:59 active-response/bin/firewall-drop: Starting
2026/03/24 04:33:59 active-response/bin/firewall-drop: {"version":1,"origin"
```

```
tirthak@wazuh-server: ~
tirthak@wazuh-server:~$ sudo iptables -L INPUT -n | grep 192.168.1
DROP       all  --  192.168.1.7          0.0.0.0/0
tirthak@wazuh-server:~$
```